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Neurocognitive outcomes and functional independence in adult survivors of childhood medulloblastoma diagnosed over 3 decades

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Abstract

Background. Treatment of childhood medulloblastoma has evolved to reduce neurotoxicity while improving survival. However, the impact of evolving therapies on late neurocognitive outcomes and adult functional independence remains unknown.

Methods. Adult survivors of childhood medulloblastoma ($n = 505$; median [minimum–maximum] age, 29 [18–46] years) and sibling controls ($n = 727$; 32 [18–58] years) from the Childhood Cancer Survivor Study completed surveys assessing neurocognitive problems and chronic health conditions (CHCs). Treatment exposures were categorized as historical (craniospinal irradiation [CSI] ≥ 30 Gy, no chemotherapy), standard-risk (CSI > 0 to <30 Gy + chemotherapy) and high-risk (CSI ≥ 30 Gy + chemotherapy) therapy. Latent class analysis identified patterns of functional independence using employment, independent living, assistance with routine/personal care needs, driver's license, and marital/partner status. Multivariable models estimated the risk of neurocognitive impairment in survivors versus siblings and by treatment exposure group, and associations between neurocognitive impairment, CHCs, and functional independence.

Results. Survivors in each treatment exposure group had a 4- to 5-fold elevated risk of impaired memory and task efficiency compared to siblings. Contemporary risk-based therapies did not confer lower risk compared to historical therapy. Survivors treated in the 1990s had a higher risk of memory impairment (relative risk [RR] 2.24, 95% confidence interval 1.39–3.60) compared to survivors treated in the 1970s. Sensorimotor, hearing problems, and seizures were associated with 33–34%, 25–26%, and 21–42% elevated risk of task efficiency and memory impairment, respectively. Treatment-related CHCs and neurocognitive impairment were associated with nonindependence.

Conclusions. Despite treatment changes, long-term survivors of childhood medulloblastoma remain at risk for neurocognitive impairment, which was associated with CHCs. Neurocognitive surveillance after contemporary regimens is imperative.

Key Points

- Adult survivors of medulloblastoma have up to a 5-fold risk of cognitive problems.
- Risk-adapted (vs. historical) therapy has not mitigated the risk of cognitive problems.
- Treatment-related chronic health conditions increase the risk of cognitive deficits.

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Importance of the Study

Previous research has demonstrated the benefits of risk-adapted therapies for childhood medulloblastoma on neurocognitive function in pediatric survivors. In adult survivors diagnosed between 1970 and 1999, approximately 60% reported task efficiency (ie, attention and processing speed) and memory impairments. Despite therapeutic changes, neither high-risk nor standard-risk contemporary therapy was associated with a lower risk of neurocognitive impairment compared to historical therapy. Furthermore, survivors treated in the 1990s were more likely to report neurocognitive problems compared to those treated in the 1970s, possibly due to the impact of chronic health conditions. Indeed, hearing deficits and a history of seizure/epilepsy conferred a 20–40% elevated risk for task efficiency and memory impairment. Moreover, these conditions and task efficiency impairment were associated with functional nonindependence. Survivorship care is crucial to prevent and manage the late effects of childhood medulloblastoma. Neurocognitive surveillance is imperative to elucidate the long-term effects of contemporary therapies on functional outcomes.

Medulloblastoma is the most common malignant central nervous system (CNS) tumor in childhood.[1] Medical advances have dramatically improved survival rates and over 80% of children diagnosed < 18 years of age will survive at least 5 years after treatment.[2] However, long-term survivors of childhood medulloblastoma experience significant neurocognitive impairments and a high burden of chronic disease that can negatively affect their socioeconomic outcomes and overall quality of life.[3–9] Among CNS tumors, survivors of medulloblastoma often have the poorest health outcomes.[10]

Treatment of medulloblastoma requires a multimodal approach, including maximal surgical resection, craniospinal irradiation (CSI), and adjuvant chemotherapy.[11] Treatment regimens have changed over time to reduce neurocognitive morbidity while maintaining or improving survival.[12,13] Historical therapy combining surgery with 30–40 Gy CSI and a local boost to the tumor bed has progressively been replaced with risk-adapted therapies that include adjuvant chemotherapy for high-risk patients and reduced-dose CSI with chemotherapy for standard-risk patients.[14] Short-term benefits of risk-adapted therapies on neurocognitive functioning have been demonstrated in few studies,[12,13,15] but little is known about long-term effects. A recent report from the Childhood Cancer Survivor Study (CCSS) found that adult survivors of medulloblastoma treated in more recent decades had reduced use of special education services compared to survivors treated with more intensive therapies,[14] suggesting a potential positive impact of cognitive abilities that underpin scholastic performance during childhood. However, data also indicate that more contemporarily treated survivors of medulloblastoma experience a higher burden of severe to life-threatening chronic conditions during adulthood,[14,16] which may, in turn, have detrimental effects on neurocognitive functioning.[17,18] Overall, it remains to be established whether evolving therapies for childhood medulloblastoma have improved long-term neurocognitive outcomes for adult survivors.

Disrupted attainment of adult developmental milestones is also of concern in long-term survivors of childhood medulloblastoma, who are less likely to graduate from college, marry, and live independently compared to their siblings.[4,19] In a past study with survivors of various childhood CNS tumor diagnoses, exposure to CSI (ie, the backbone of medulloblastoma treatment) was associated with 320% higher odds of being socially and functionally nonindependent compared to no exposure to cranial irradiation.[20] However, patterns of functional independence have not been examined in a homogenous sample of medulloblastoma survivors, and the contribution of neurocognitive late effects and chronic health conditions to the ultimate outcome of independence have not been investigated in this population.

Therefore, the aims of this study were to investigate the effects of evolving therapies for childhood medulloblastoma on neurocognitive functioning and to examine the associations between neurocognition, physical health morbidities, and functional independence in adult survivors.

Materials and Methods

Participants

The CCSS is a retrospective cohort study with prospective follow-up based on a survey that involves 5-year survivors of childhood cancer diagnosed before the age of 21 years between 1970 and 1999 and treated at one of the 31 participating institutions in North America.[21] The current study included survivors aged 18 or older with self- or proxy-completed neurocognitive questionnaires at follow-up 2 or follow-up 5. Survivors who reported having a genetic syndrome associated with neurocognitive impairment but unrelated to their primary cancer diagnosis (eg, Klinefelter's or Turner's syndrome) were excluded. The Institutional Review Board at each participating institution approved the CCSS study protocol, and all participants gave written informed consent prior to any research procedures.

Treatment Exposures

Diagnosis and treatment history (within the first 5 years of diagnosis) were abstracted from medical records. To examine the evolving therapies for childhood medulloblastoma, 3 treatment exposure groups, adapted

from a previous CCSS study,[14] were defined: historical therapy (surgery + CSI ≥ 30 Gy, without chemotherapy), high-risk multimodal therapy (surgery + CSI ≥ 30 Gy + chemotherapy), and standard-risk multimodal therapy (surgery + CSI > 0 to <30 Gy + chemotherapy). Individual treatment exposures of CSI (>0 to <30 Gy vs. ≥30 Gy) and chemotherapy (yes vs. no) were also examined. Three treatment era groups were also defined based on decade of diagnosis: 1970–1979, 1980–1989, and 1990–1999.

Outcomes

Neurocognitive functioning was evaluated through the CCSS Neurocognitive Questionnaire.[22,23] This questionnaire has been validated among childhood cancer survivors to examine the perception of neurocognitive disturbances in relation to daily functioning. A 3-point Likert scale was used to rate the significance of neurocognitive problems experienced by survivors in the domains of task efficiency (ie, processing speed and attention), emotional regulation (ie, control of emotions and frustration tolerance), organization (of materials, environments and activities), and memory (ie, short-term and long-term). Impairment was defined as scores ≥ 90th percentile of sibling norms.

Patient-reported chronic health conditions were classified and graded using the National Cancer Institute's Common Terminology Criteria for Adverse Events version 4.3.[24] Eight organ systems were selected a priori given their potential impact on neurocognitive function[17,18]: sensorimotor, hearing, vision, stroke, seizure/epilepsy, endocrine, pulmonary, and cardiac. The maximum grade for each organ system was dichotomized as 0–1 (none or mild) versus 2–4 (moderate, severe, or life-threatening). Of note, grade 1 sensory neuropathy was added to the sensorimotor conditions as this is the maximum grade assigned to this condition in CCSS.

Functional independence was examined via 5 indicators of financial autonomy, social partnership, and activities of daily living used in previous research with brain tumor survivors[20]: (1) employment (full time [working full time, caring for home or family, student], part time [working part time], or unemployed [not currently working, unemployed and looking for work, unable to work due to disability or illness, retired]); (2) marital/partner status (never married [single, never married], or history of marriage [married or no longer married, living as married or no longer living as married, widowed, divorced, separated]); (3) independent living (yes [lives with spouse/partner or alone] or no [lives with parents or other relatives]); (4) current driver's license (yes or no); and (5) assistance with personal care needs and/or routine needs (such as eating and dressing, or everyday chores and shopping; yes or no).

Statistical Analyses

Descriptive statistics were presented for all predictors and outcomes. Multivariable generalized equation models compared the risk of neurocognitive impairment in survivors and sibling controls, adjusting for sex, race/ethnicity, and age at survey completion. Multivariable modified Poisson regression models evaluated the risk of neurocognitive impairment associated with treatment exposure groups and the presence of grade ≥ 2 chronic health conditions, adjusting for sex, race/ethnicity, age at survey completion, age at diagnosis, and relapse/second neoplasm. The interactions between treatment exposure groups and treatment era, and between chronic conditions and treatment era, on neurocognitive outcomes were tested and not included in the models because they were not statistically significant. Relative risk (RR) and 95% confidence interval (CI) were reported.

The 5 indicators of functional independence were used to identify data-driven classes through a Latent Class Analysis (LCA). LCA is a statistical method used to identify unobserved groups (ie, latent classes) within a population based on patterns of responses to observed variables. Instead of directly observing which group each individual belongs to, LCA estimates the probability of each individual belonging to each latent class. Because the achievement of adulthood milestones requires time, functional independence was examined only among survivors ≥ 25 years of age. Models were fitted with 1–5 classes, and the final model was selected based on clinical meaning and multiple fit indices, including adjusted Bayesian information criterion and entropy. Multivariable modified Poisson regression models evaluated the associations of treatment exposure groups, neurocognitive impairment, and chronic health conditions with classes of functional independence, adjusting for sex, race/ethnicity, age at diagnosis, and time since diagnosis. SAS v9.4 (SAS Institute) was used for the analyses.

Results

Sample Characteristics

Of the 576 eligible medulloblastoma survivors who completed the baseline CCSS survey, 505 were evaluable for the current study (Supplementary Figure 1; Table 1[25]). Survivors (58% male) had a median age of 7 years (minimum–maximum, 0–20 years) at diagnosis and a median age of 29 years (minimum–maximum, 18–46 years) at survey completion. Among these 505 survivors, 98 (19%) were treated with historical therapy, 120 (24%) with standard-risk multimodal therapy, and 186 (37%) with high-risk multimodal therapy. The remaining 101 survivors (20%) had missing exposure information ($n = 48$) or received other treatment combinations (ie, no CSI [$n = 31$] or CSI > 0 to <30 Gy without chemotherapy [$n = 22$]) and were excluded from subsequent analyses that included treatment exposures.

Historical therapy was more likely to be used with survivors treated in the 1970s compared to those treated in the 1980s or 1990s, whereas the use of standard-risk multimodal therapy increased over the 3 decades (all P's < .001; Supplementary Table 1). The use of high-risk multimodal therapy did not vary significantly by decade ($P > .05$). Over the 3 decades, fewer survivors received ≥30 Gy CSI (77%, 71% and 51%), while more survivors received no CSI (3%,

Table 1. Demographic and Clinical Characteristics of Medulloblastoma Survivors by Treatment Exposure Group and Sibling Controls

Characteristics	Historical[a] (*n* = 98) *n* (%)	Standard-Risk Multimodal[b] (*n* = 120) *n* (%)	High-Risk Multimodal[c] (*n* = 186) *n* (%)	Siblings (*N* = 727) *n* (%)
Median (min–max) age at survey completion, years	32 (20–46)	26 (18–39)	30.5 (18–40)	32 (18–58)
Age at diagnosis, years				
<3 years	5 (5.1%)	15 (12.5%)	12 (6.5%)	/
≥3 years	93 (94.9%)	105 (87.5%)	174 (93.5%)	/
Median (min–max)	8 (2–18)	6 (0–18)	8 (1–19)	/
Sex				
Male	51 (52.0%)	64 (53.3%)	128 (68.8%)	318 (43.7%)
Female	47 (48.0%)	56 (46.7%)	58 (31.2%)	409 (56.3%)
Race/ethnicity				
White, non-Hispanic	81 (82.7%)	102 (85.0%)	161 (86.6%)	652 (89.7%)
Black, non-Hispanic	5 (5.1%)	4 (3.3%)	10 (5.4%)	13 (1.8%)
Hispanic	2 (2.0%)	8 (6.7%)	8 (4.3%)	25 (3.4%)
Other	10 (10.2%)	6 (5.0%)	7 (3.8%)	37 (5.1%)
Chronic health conditions (grade ≥ 2)[d]				
Sensorimotor	45 (45.9%)	57 (47.5%)	94 (50.5%)	/
Hearing[e]	33 (33.7%)	62 (51.7%)	95 (51.1%)	/
Vision	6 (6.1%)	9 (7.5%)	20 (10.8%)	/
Stroke [e]	7 (7.1%)	14 (11.7%)	42 (22.6%)	/
Seizure/epilepsy	11 (11.2%)	13 (10.8%)	29 (15.6%)	/
Endocrine[e]	57 (58.2%)	89 (74.2%)	145 (78.0%)	/
Pulmonary	6 (6.1%)	8 (6.7%)	11 (5.9%)	/
Cardiac	14 (14.3%)	9 (7.5%)	22 (11.8%)	/
Relapse/subsequent neoplasm	15 (15.3%)	12 (10.0%)	21 (11.3%)	/
Treatments				
Surgery	98 (100.0%)	120 (100.0%)	186 (100.0%)	/
Any chemotherapy	/	120 (100.0%)	186 (100.0%)	/
Craniospinal irradiation				/
>0 to <30 Gy	/	120 (100.0%)	/	
≥30 Gy	98 (100.0%)	/	186 (100.0%)	
Median (min–max) dose, Gy	35 (30–45)	23 (0.2–29)	36 (30–48)	/
Cisplatin, mg/m^2				/
None	/	23 (19.7%)	76 (41.8%)	/
>0 to <200	/	6 (5.1%)	9 (4.9%)	/
200 to <400	/	46 (39.3%)	52 (28.6%)	/
≥400	/	42 (35.9%)	45 (24.7%)	/
Median (min–max) dose	/	377.05 (150.59–830.00)	342.39 (79.59–1358.49)	/
Alkylating agents, mg/m^2[f]				
None	/	8 (7.1%)	26 (14.9%)	/
>0 to <4000	/	7 (6.3%)	12 (6.9%)	/
4000 to <8000	/	17 (15.2%)	22 (12.6%)	/
≥8000	/	80 (71.4%)	114 (65.5%)	/
Median (mix–max) dose	/	9839.68 (1165.05–65709.30)	9895.23 (9.60–33931.03)	/

Table 1. Continued

Characteristics	Historical[a] (*n* = 98) *n* (%)	Standard-Risk Multimodal[b] (*n* = 120) *n* (%)	High-Risk Multimodal[c] (*n* = 186) *n* (%)	Siblings (*N* = 727) *n* (%)
Carboplatin, mg/m^2	/	9 (7.5%)	32 (17.2%)	/
Median (min–max) dose	/	1017.70 (696.11–2001.16)	1790.00 (265.01–8310.42)	/
Cyclophosphamide	/	59 (49.2%)	82 (44.1%)	/
Lomustine	/	53 (44.2%)	90 (48.4%)	/
Vincristine	/	112 (93.3%)	163 (87.6%)	/
Etoposide	/	15 (12.5%)	51 (27.4%)	/

Abbreviations: max, maximum; min, minimum. One hundred one survivors who had missing treatment exposure information (*n* = 48), or received other treatment combinations (ie, no craniospinal irradiation [*n* = 31] or craniospinal irradiation > 0 to <30 Gy without chemotherapy [*n* = 22]) were excluded from the primary analyses involving treatment exposures. Out of these 101 survivors, 18 were diagnosed before 3 years of age and did not receive craniospinal irradiation. Dashes indicate clinical variables that were not applicable to sibling controls or treatment exposure groups.
[a]Historical therapy: surgery + craniospinal irradiation ≥ 30 Gy, no chemotherapy.
[b]Standard risk multimodal therapy: surgery + craniospinal irradiation > 0 to <30 Gy + chemotherapy.
[c]High-risk multimodal therapy: surgery + craniospinal irradiation ≥ 30 Gy + chemotherapy.
[d]Common Terminology Criteria for Adverse Events grading system: 1 = mild, 2 = moderate, 3 = severe/disabling, 4 = life threatening. Pulmonary conditions also included grade 1 chronic cough requiring oxygen because grade 1 is the maximum grade for this condition. Neurologic conditions also included grade 1 sensory neuropathy because all sensory neuropathy conditions in CCSS are considered grade 1.
[e]Statistically significant difference in the prevalence of hearing problems, stroke, and endocrine conditions across the 3 treatment exposure groups (all *P*'s < .05). For hearing problems and endocrine conditions, historical therapy is statistically different from standard-risk and high-risk therapies (*P*'s < .05). For stroke, historical and standard-risk therapies are statistically different from high-risk therapy (*P*'s < .05).
[f]Cyclophosphamide-equivalent dose, calculated using the method described by Green et al.[25].

Table 2. Neurocognitive Impairment in Medulloblastoma Survivors Compared to Sibling Controls

Groups	Task Efficiency RR (95% CI)	Emotional Regulation RR (95% CI)	Organization RR (95% CI)	Memory RR (95% CI)
Sibling controls	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
Historical	**4.09 (3.21–5.22)**	**2.13 (1.36–3.34)**	**1.65 (1.13–2.40)**	**4.61 (3.37–6.30)**
Standard-risk	**4.19 (3.27–5.38)**	**3.04 (1.98–4.66)**	**1.77 (1.19–2.63)**	**4.32 (3.12–5.96)**
High-risk	**4.26 (3.40–5.33)**	**3.18 (2.20–4.59)**	**2.46 (1.85–3.28)**	**5.31 (4.06–6.93)**

Multivariable models were further adjusted for sex, race/ethnicity, and age at survey completion. Boldface indicates statistical significance.

4% and 10%) or >0 to <30 Gy CSI (20%, 25% and 39%; Supplementary Table 1; all *P*'s < .05). The median CSI dose across treatment decades slightly decreased over time (Supplementary Table 1).

Neurocognitive Impairment

Across all treatment exposure groups and decades of diagnosis, more than 60% of survivors were impaired in at least 1 neurocognitive domain (Supplementary Table 2). More than two-thirds (71%) of the neurocognitive questionnaires were self-completed by survivors. The prevalence of impairment by self- and proxy-completed questionnaires is shown in Supplementary Table 3. After adjusting for sex, race/ethnicity, and age at assessment, survivors in each treatment exposure group had a significantly elevated risk of neurocognitive impairment compared to sibling controls, with the highest risk observed in the domains of task efficiency (eg, high-risk: RR 4.26, 95% CI 3.40–5.33) and memory (eg, high-risk: RR 5.31, 95% CI 4.06–6.93; Table 2).

Among survivors, there was a similar risk of impairment in the contemporary risk-based therapy groups compared to those treated with historical therapy (Table 3). Nonetheless, survivors who received high-risk multimodal therapy had elevated risk of organization impairment compared to survivors of standard-risk therapy (RR 1.50, 95% CI 1.01–2.23; Supplementary Table 4), after adjusting for sex, race/ethnicity, age at survey completion, age at diagnosis, and relapse/subsequent neoplasm. This effect remained significant when further adjusting for the treatment era (Supplementary Table 4). In models including both treatment exposure groups and a decade of diagnosis, survivors treated in the 1990s were more likely to report organization (RR 2.83, 95% CI 1.44–5.57) and memory (RR 2.24, 95% CI 1.39–3.60) impairments compared to survivors treated in the

1970s (Table 3 and Supplementary Table 4). For each 1-year increase in age at diagnosis, survivors had 5–6% lower risk of task efficiency impairment (Table 3). Compared to their male counterparts, female survivors had 24–27% increased risk of task efficiency and memory impairment, and 62–64% increased risk of emotional regulation and organization impairment (Table 3). In models examining individual treatment exposures instead of treatment exposure groups, CSI ≥ 30 Gy (but not chemotherapy) was associated with neurocognitive impairment, specifically in the domain of organization (Supplementary Table 5).

Neurocognitive Impairment and Chronic Health Conditions

Several grades 2–4 chronic health conditions were found to be associated with neurocognitive outcomes, after adjusting for sex, race/ethnicity, age at survey completion, age at diagnosis, and relapse/subsequent neoplasm (Table 4). Sensorimotor conditions were associated with 33–93% elevated risk of neurocognitive impairment in each domain. Hearing problems were associated with a higher risk of impaired task efficiency (RR 1.25, 95% CI 1.08–1.44) and memory (RR 1.26, 95% CI 1.04–1.53); similar associations were observed for seizure/epilepsy (impaired task efficiency: RR 1.21, 95% CI 1.03–1.42; impaired memory: RR 1.42, 95% CI 1.13–1.77). Pulmonary conditions were associated with emotional regulation impairment (RR 1.82, 95% CI 1.22–2.71).

Classes of Functional Independence

The analyses of functional independence included 394 survivors (78%) who were aged 25 years or older. The 3-class model demonstrated the best fit to the data (Supplementary Table 6). Survivors were classified as independent (37%), moderately independent (ie, nonindependent living and never married; 21%) and nonindependent (42%). Figure 1 and Supplementary Table 7 show the distribution of the 5 individual indicators by class of functional independence. Nonindependent survivors had a higher prevalence of sensorimotor, hearing, vision, and neurologic conditions (all P's < .01; Supplementary Table 8). The prevalence of nonindependent survivors was higher in the high-risk therapy group compared to the other 2 treatment exposure groups ($P = .012$), whereas the distribution of survivors in each functional independence class did not differ by treatment era (Supplementary Table 9).

In multivariate models adjusted for sex, race/ethnicity, age at diagnosis, and time since diagnosis, there were no significant associations between the treatment exposure groups and the classes of functional independence (Table 5). No significant associations were found in models considering individual treatment exposures of CSI and chemotherapy (Supplementary Table 10). Survivors with impaired task efficiency had an elevated risk of being nonindependent (RR 1.93, 95% CI 1.42–2.64; Table 5). Survivors with hearing impairment (RR 1.28, 95% CI 1.05–1.56) and seizure/epilepsy (RR 1.54, 95% CI 1.25–1.89) were at elevated risk for functional nonindependence. There

Table 3. Associations Between Treatment Exposures and Neurocognitive Impairment Among Medulloblastoma Survivors Using the Historical Therapy Group as Referent

Model/Predictor and Category	Task Efficiency	Emotional Regulation	Organization	Memory
	RR (95% CI)	RR (95% CI)	RR (95% CI)	RR (95% CI)
Model 1				
Treatment exposure group				
Historical	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
Standard-risk	0.92 (0.74–1.14)	0.71 (0.42–1.22)	0.95 (0.57–1.56)	1.09 (0.79–1.50)
High-risk	1.00 (0.81–1.22)	1.05 (0.67–1.63)	1.42 (0.93–2.15)	1.23 (0.93–1.63)
Age at diagnosis	**0.95 (0.92–0.97)**	1.00 (0.93–1.06)	0.97 (0.92–1.02)	1.01 (0.97–1.04)
Female sex (vs. male)	**1.27 (1.08–1.48)**	**1.63 (1.14–2.35)**	**1.64 (1.20–2.24)**	**1.25 (1.01–1.54)**
Model 2				
Treatment exposure group				
Historical	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
Standard-risk	0.88 (0.70–1.10)	0.59 (0.34–1.03)	0.79 (0.48–1.32)	0.92 (0.67–1.26)
High-risk	0.97 (0.78–1.19)	0.93 (0.59–1.46)	1.20 (0.79–1.84)	1.07 (0.81–1.41)
Treatment era				
1970–1979	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
1980–1989	0.99 (0.73–1.33)	0.74 (0.37–1.50)	1.54 (0.80–2.99)	1.27 (0.80–2.02)
1990–1999	1.14 (0.83–1.55)	1.39 (0.69–2.78)	**2.83 (1.44–5.57)**	**2.24 (1.39–3.60)**
Age at diagnosis	**0.94 (0.91–0.97)**	0.97 (0.90–1.04)	**0.91 (0.86–0.97)**	0.96 (0.92–1.00)
Female sex (vs. male)	**1.26 (1.08–1.48)**	**1.62 (1.13–2.32)**	**1.64 (1.21–2.23)**	**1.24 (1.01–1.51)**

Multivariable models were further adjusted for race/ethnicity, age at survey completion, and relapse/subsequent neoplasm. Model 1 includes treatment exposure groups. Model 2 includes treatment exposure groups and treatment era. Boldface indicates statistical significance.

Table 4. Associations Between Chronic Health Conditions and Neurocognitive Impairment Among Medulloblastoma Survivors

Chronic Health Conditions[a]	Task Efficiency	Emotional Regulation	Organization	Memory
	RR (95% CI)	RR (95% CI)	RR (95% CI)	RR (95% CI)
Sensorimotor	**1.33 (1.14–1.54)**	**1.93 (1.35–2.75)**	**1.85 (1.36–2.49)**	**1.34 (1.10–1.63)**
Hearing	**1.25 (1.08–1.44)**	1.01 (0.73–1.38)	1.06 (0.80–1.40)	**1.26 (1.04–1.53)**
Vision	0.91 (0.70–1.19)	0.98 (0.58–1.64)	1.10 (0.69–1.77)	0.95 (0.68–1.31)
Stroke	0.91 (0.74–1.11)	0.96 (0.62–1.49)	0.97 (0.66–1.44)	0.93 (0.71–1.22)
Seizure/epilepsy	**1.21 (1.03–1.42)**	1.42 (0.95–2.11)	0.96 (0.65–1.41)	**1.42 (1.13–1.77)**
Endocrine	1.00 (0.84–1.18)	0.96 (0.67–1.38)	1.00 (0.73–1.37)	1.23 (0.97–1.55)
Pulmonary	1.10 (0.89–1.37)	**1.82 (1.22–2.71)**	1.34 (0.89–2.01)	1.24 (0.95–1.63)
Cardiac	1.07 (0.86–1.32)	1.21 (0.80–1.82)	0.94 (0.62–1.44)	0.92 (0.69–1.23)
Age at diagnosis	**0.95 (0.92–0.97)**	1.00 (0.95–1.06)	0.96 (0.92–1.01)	1.01 (0.98–1.05)

[a]Chronic health conditions were classified and graded in severity using the Common Terminology Criteria for Adverse Events grading system: 1 = mild, 2 = moderate, 3 = severe/disabling, 4 = life-threatening. Pulmonary conditions also included grade 1 chronic cough requiring oxygen as grade 1 is the maximum grade for this condition. Neurologic conditions also included grade 1 sensory neuropathy because all sensory neuropathy conditions in CCSS are considered grade 1.
Multivariable models were further adjusted for sex, race/ethnicity, age at survey completion, and relapse/subsequent neoplasm. Separate models for each neurocognitive outcome; all chronic health conditions in each model. Boldface indicates statistical significance.

were no significant associations between neurocognitive impairment or chronic health conditions and moderate functional independence (Table 5). Across models, sex was not associated with failure to attain full independence, whereas each 1-year increase in age at diagnosis was associated with up to 10% lower risk of functional nonindependence and each 1-year increase in time since diagnosis was associated with a 6% higher probability of being fully independent (Table 5).

Discussion

In a cohort of over 500 adult survivors of childhood medulloblastoma diagnosed between 1970 and 1999, we examined the potential impact of evolving therapies on neurocognitive functioning and attainment of functional independence in adulthood. The prevalence of neurocognitive impairment was high, with over 60% of our sample reporting problems in at least 1 area of functioning. Treatment with any therapeutic regimen was associated with up to 5-fold elevated risk of neurocognitive impairment compared to sibling controls. Surprisingly, modern risk-based treatment approaches showed few benefits on neurocognitive functioning compared to historical therapy, and survivors treated in the most recent decade were more likely to report neurocognitive impairment. Almost two-thirds of survivors did not achieve full independence as adults, which was associated with neurocognitive, neurosensory, and neurologic morbidities. Overall, these results highlight that, despite changes in front-line therapies, long-term survivors of childhood medulloblastoma remain at risk for long-term neurocognitive problems and physical health morbidities that significantly disrupt their ability to attain social and functional independence in adulthood.

Historical therapy for childhood medulloblastoma combining surgery and high-dose CSI has been adapted to reduce CSI dose and volume and incorporate novel chemotherapy agents based on patients' risk status.[14] As expected, in our cohort of survivors diagnosed over 3 decades, we observed a decrease in treatment with historical therapy (ie, high-dose CSI and no chemotherapy) and a concurrent increase in treatment with risk-based therapies. In addition, the use of CSI ≥ 30 Gy decreased over this time, but the median CSI dose was only marginally reduced. Despite these therapeutic changes, there were no differences in the risk of neurocognitive problems in survivors treated with standard-risk or high-risk therapy versus historical therapy. It is possible that survivors of historical therapy experienced poor health outcomes and did not survive long enough to participate in the study. Alternatively, survivors of historical therapy, most of whom were treated in the 1970s, may have adapted to their deficits over time and applied different internal standards for evaluating their neurocognitive functioning.[26] Indeed, in models adjusted for the treatment era, there was a higher risk of neurocognitive impairment in survivors treated in the 1990s compared to the 1970s. In a previous study with the CCSS cohort, medulloblastoma survivors treated in the 1990s demonstrated 44% and 189% higher risk of reporting at least 1 or multiple severe-to-fatal chronic conditions, respectively, compared to those treated in the 1970s.[14] Therefore, the intensification of therapy for high-risk patients and the higher burden of late-onset chronic conditions experienced by medulloblastoma survivors treated in more recent decades could explain their elevated risk of neurocognitive deficits, given the negative impact of chronic health conditions on neurocognitive function in childhood cancer survivors,[17] especially those exposed to CNS-directed therapies.[18]These results support prioritizing long-term neurocognitive surveillance in medulloblastoma survivors, even in those recently treated, and underscore

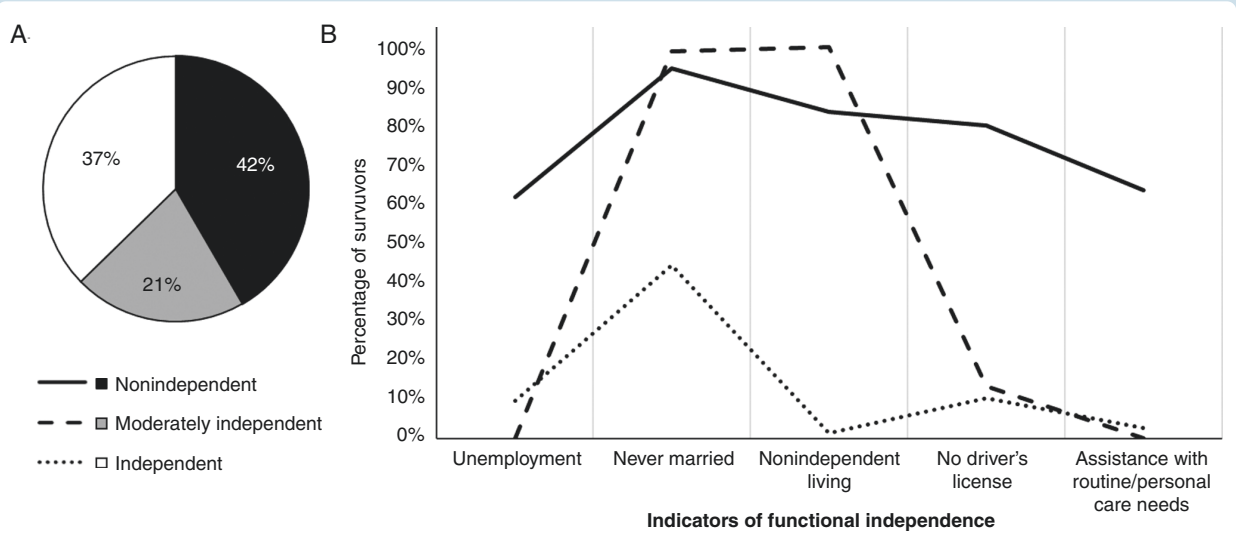

Figure 1. Classes of functional independence among adult survivors of childhood medulloblastoma, obtained through latent class analysis. (A) Percentage of the whole sample categorized in each class of functional independence. (B) Percentage of survivors with a negative outcome on the 5 indicators of functional independence in each class.

the need to continue to reduce treatment-related toxicity while improving or maintaining survival.

The comparison between modern risk-adapted therapies demonstrated that survivors treated with standard-risk therapy reported fewer neurocognitive problems than those treated with high-risk therapy only in the domain of organization. Similarly, survivors treated with less than 30 Gy CSI showed less organization impairment compared to those treated with 30 Gy or more. Together, these results suggest that the early advantage of de-escalated therapies (especially reduced CSI dose) previously reported among pediatric survivors[12,13,15] may be only partially maintained in adulthood. Several factors could account for the poor outcomes observed in the current study with adult survivors, including longer time since diagnosis[8] and the use of patient-reported outcomes indicative of day-to-day functional concerns.[22,23] Nonetheless, the overall risk of neurocognitive impairment in survivors remains indisputable. Even treatment with standard-risk therapy[13,15] was associated with up to 332% elevated risk of neurocognitive impairment compared to the sibling counterparts. Overall, the highest risk was observed in the domains of memory and task efficiency, reflecting problems with working memory, attention and processing speed.[22,23] Within heterogeneous samples of CNS tumor survivors in the CCSS cohort, task efficiency and memory were the domains most frequently impaired at baseline[3] and most likely to emerge as new impairments in previously unimpaired adult survivors at 10-year follow-up.[27] Because increasing numbers of children diagnosed with medulloblastoma survive beyond 5 years, these results are indicative of a concerning trend of a growing population of medulloblastoma survivors who face a significant risk of neurocognitive deficits as they age.

As an increasing number of medulloblastoma survivors reach adulthood, the attainment of functional independence has become a primary concern.[20] In this study, over 60% of medulloblastoma survivors failed to achieve complete independence as adults. One previous report with adult survivors treated for a variety of CNS tumors identified similar classes of functional independence.[20] It also found that 26% of the sample was functionally nonindependent, with CSI being the most significant risk factor compared to no cranial radiation exposure.[20] In our study, a larger proportion of survivors were functionally nonindependent (42%) arguably due to the exposure of the whole sample to CSI. In addition, reduced CSI dose (either examined independently or conceptualized as standard-risk therapy) was not associated with functional independence, suggesting no impact of CSI dose on that outcome. Because CSI remains the backbone of medulloblastoma treatment, it is vital to identify potentially modifiable risk factors of independence that are amenable to intervention. In the aforementioned study with CNS tumor survivors, physical strength deficits (but not IQ impairment) conferred a higher risk of being moderately independent,[20] whereas no neurocognitive or physical health morbidities were significantly associated with moderate independence in our sample restricted to medulloblastoma survivors. We recently demonstrated that social cognitive skills (ie, perception and reasoning from social stimuli) are particularly impaired in medulloblastoma survivors, and better social cognition uniquely predicted marital status and independent living beyond the impact of neurocognitive (ie, executive function) impairment.[28] Because moderately independent survivors of childhood medulloblastoma reported negative outcomes almost exclusively on these socially relevant indicators (ie, independent living and marital status), facial affect recognition and social cue processing may be novel intervention targets to investigate in order to specifically address the domains of concern in these survivors. Moderately independent survivors were also closer in time to their primary diagnosis, and further follow-up assessment will be important to elucidate

Table 5. Treatment Exposure Groups, Neurocognitive Impairment, Chronic Health Conditions and the Functional Independence in Medulloblastoma Survivors

Model/Predictor and Category	Moderate Functional Independence	Functional Nonindependence
	RR (95% CI)	RR (95% CI)
Model 1: Treatment exposure group		
Standard-risk	1.00 (referent)	1.00 (referent)
Historical	0.89 (0.55–1.43)	1.00 (0.72–1.39)
High-risk	0.88 (0.56–1.39)	1.28 (0.99–1.65)
Age at diagnosis	**0.86 (0.81–0.91)**	**0.90 (0.86–0.93)**
Time since diagnosis	0.94 (0.89–1.00)	0.99 (0.96–1.02)
Female sex (vs. male)	0.93 (0.63–1.37)	1.04 (0.85–1.27)
Model 2: Neurocognitive impairment		
Task efficiency	1.32 (0.86–2.01)	**1.93 (1.42–2.64)**
Emotional regulation	0.86 (0.46–1.59)	1.07 (0.86–1.32)
Organization	1.04 (0.65–1.66)	1.13 (0.92–1.38)
Memory	0.75 (0.48–1.18)	0.81 (0.66–1.00)
Age at diagnosis	**0.86 (0.81–0.92)**	**0.92 (0.89–0.96)**
Time since diagnosis	**0.94 (0.89–1.00)**	0.99 (0.96–1.02)
Female sex (vs. male)	0.93 (0.63–1.37)	0.89 (0.73–1.08)
Model 3: Chronic health conditions		
Sensorimotor	0.87 (0.56–1.33)	1.24 (1.00–1.53)
Hearing	1.25 (0.85–1.85)	**1.28 (1.05–1.56)**
Vision	1.08 (0.42–2.77)	1.20 (0.94–1.52)
Stroke	0.97 (0.48–1.95)	1.17 (0.94–1.45)
Seizure/epilepsy	1.28 (0.63–2.59)	**1.54 (1.25–1.89)**
Endocrine	0.91 (0.61–1.37)	**0.77 (0.61–0.97)**
Pulmonary	0.35 (0.06–2.14)	1.14 (0.85–1.55)
Cardiac	0.99 (0.52–1.90)	1.20 (0.92–1.58)
Age at diagnosis	**0.85 (0.80–0.91)**	**0.90 (0.87–0.94)**
Time since diagnosis	**0.94 (0.88–0.99)**	0.99 (0.96–1.02)
Female sex (vs. male)	0.96 (0.65–1.41)	0.98 (0.80–1.19)

Multivariable models were further adjusted for race/ethnicity. Separate models were used for each class of independence; functional independence class was considered the reference group. Boldface indicates statistical significance.

changes in their functional independence status later in survivorship. Additional medical procedures (eg, shunt placement[20]) as well as physical health (eg, frail physiologic status[29]) and psychosocial factors (eg, depression[30]) that may contribute to functional independence also warrant further investigation.

Aging survivors of pediatric CNS malignancies, and in particular medulloblastoma, experience a substantial burden of chronic health conditions.[14,16] In our cohort, we demonstrated that moderate to life-threatening chronic health conditions, especially neurosensory and neurologic conditions, contribute to neurocognitive impairment as well as nonindependence in adult survivors. Seizures may negatively affect neurocognitive outcomes through ion channel alteration, inflammation, or reduced neuroplasticity,[31] whereas hearing loss may result in structural cortical reorganization or cognitively effortful listening due to missing or deteriorated auditory input.[32] Furthermore, these conditions significantly limit the ability to achieve adult social milestones such as full-time employment and independent living arrangement in clinical populations.[33] Neurologic problems[15] and hearing loss[34] have been associated with poor intellectual outcomes and worsening neurocognitive function over time in pediatric samples of medulloblastoma survivors. The few studies that examined these associations among adult survivors used heterogenous samples of CNS tumors and considered specific chronic health conditions and social attainment outcomes in isolation.[8,27,35,36] The current study expands these past findings by demonstrating concurrent associations among neurosensory and neurologic conditions and neurocognitive impairment with the ultimate outcome of functional independence in adult survivors of childhood medulloblastoma.

While the elevated risk of neurologic and psychosocial problems has been reported among adult survivors treated

for childhood medulloblastoma/PNET,[19] our findings indicate that such neurologic and neurosensory conditions may play a role in the failure to achieve independence in medulloblastoma survivors, thus interventions targeting these health conditions may be viable targets to improve independence. For example, hearing aids can significantly reduce cognitive decline[37] and improve participation in social activities[38] and instrumental activities of daily living[39] in the general population, but hearing loss remains an overlooked "invisible disability" in childhood cancer survivors.[40] Seizure resolution is associated with a lower probability of being single or never married in adult survivors of childhood CNS tumors.[36] Lastly, behavioral remediation programs designed to achieve patient-specific meaningful functional goals have the potential to improve cognitive and real-life functioning in brain tumor survivors.[41] Therefore, our study identifies novel (neurologic and neurosensory diseases) and established (neurocognitive impairment) intervention targets that may promote functional independence in long-term survivors of childhood medulloblastoma. Given the limited observed impact of therapeutic changes on long-term outcomes, prevention or remediation of neurocognitive and neurologic late effects will be crucial to promote the achievement of functional independence in adulthood.

The current findings should be considered in the context of several limitations. Compared to a previous CCSS report,[14] we used slightly modified definitions of treatment exposure groups that were not restricted to specific chemotherapy agents. This approach may not reflect most common medulloblastoma regimens, but it includes additional chemotherapy agents associated with chronic health conditions that have been shown to significantly increase the risk of neurocognitive deficits among CNS-treated survivors (eg, vincristine and peripheral neuropathy[18]). Our results also may be influenced by differences in neurocognitive impairments reported by survivors and their proxies, possibly due to different internal standards used by these respondents. However, the inclusion of proxy-completed questionnaires averted the exclusion of survivors who were presumably too impaired to complete the questionnaire independently.[3] Self-reported neurocognitive functioning is relatively easy to assess in a large, geographically dispersed population as the CCSS cohort, and it provides valuable information about day-to-day functional impairments that could inform the development of compensatory cognitive–behavioral interventions. However, the use of patient-reported measures may bias the true prevalence of neurocognitive deficits. Gold-standard neurocognitive testing and medical examinations should be used in future research with medulloblastoma survivors given the complexity of their morbidities. Because neurocognitive functioning, chronic conditions, and socioeconomic indicators of independence were collected at the same time point, causal/temporal relationships among these outcomes cannot be inferred. Baseline measures of neurocognitive functioning were not available and could not be controlled for in the analyses. A range of predictors examined in the current analysis and known risk factors, such as female sex and younger age at diagnosis, were confirmed as significant predictors for poor outcomes. However, it was not possible to assess the impact of tumor characteristics such as location and size, because they are not consistently available in the CCSS cohort. Additional risk factors could contribute to substantial morbidities observed in adult survivors, including posterior fossa syndrome/cerebellar mutism,[42] anesthesia,[43] and health behaviors.[27] Due to the small number of survivors diagnosed before the age of 3, the impact of increased systemic chemotherapy to avoid or defer CSI for infant medulloblastoma could not be investigated, and the potential advantages of this therapeutic approach warrant future research with contemporarily treated survivors. Lastly, the results of this study cannot be generalized to survivors treated in more recent decades who benefitted from modern neuroimaging methods and high-precision radiotherapy techniques, such as intensity-modulated and proton radiation therapy,[44] possibly resulting in superior intellectual outcomes.[45] However, benefits of these novel radiotherapies do not appear to extend to social outcomes (ie, social skills and peer relationships) in pediatric age,[46] therefore, it remains vital to provide neurocognitive and medical surveillance in recently treated survivors to promote the achievement of social and functional outcomes in adulthood. Meanwhile, ongoing trials further de-escalating therapies for molecular subgroups (ie, reduced CSI for WNT diseases, and reduced chemotherapy for infant SHH diseases) as well as trials integrating neuroprotective (ie, memantine) and otoprotective (ie, sodium thiosulfate) agents will hopefully yield promising results and reduce late effects in future cohorts.[47] In the molecular era of risk-adapted therapies for childhood medulloblastoma, prospective studies with long-term follow-up and repeated assessments from childhood to adulthood will be important to characterize trajectories of neurocognitive function,[48] chronic conditions, and functional independence.

In conclusion, modern risk-adapted therapies for medulloblastoma were not associated with better neurocognitive outcomes in long-term adult survivors compared to historical therapy. Survivors treated in the 1990s demonstrated a greater risk of neurocognitive impairment compared to those treated in the 1970s, which could be explained by the increasing burden of chronic health conditions in recent decades. The majority of medulloblastoma survivors in our cohort did not achieve full functional independence in adulthood and survivors treated with standard-risk therapy had a similar risk of being nonindependent as survivors treated with high-risk therapy. However, neurocognitive and neurologic late effects were associated with functional nonindependence, pinpointing modifiable risk factors that should be targeted in future interventions and novel clinical trials. Regular surveillance of neurocognitive functioning[49] in medulloblastoma survivors remains critical to elucidate the long-term effects of contemporary therapies, and the implementation of follow-up screening guidelines becomes a priority for the prevention and management of treatment-related late effects with the goal of optimizing attainment of independence in adulthood.

Supplementary material

Supplementary material is available online at *Neuro-Oncology* (https://academic.oup.com/neuro-oncology).

Keywords

childhood cancer | functional independence | medulloblastoma | neurocognitive impairment | survivorship

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Conflict of interest statement

None declared.

Authorship statement

Conceptualization: I.T.O., W.M.L., G.T.A., K.R.K., T.M.B.; Methodology: I.T.O., W.M.L., G.T.A., L.L.R., K.R.K., T.M.B.; Resources: W.M.L., R.M.H., K.C.O., G.T.A.; Formal analysis: S.M., M.X., D.S.; Writing—original draft: C.P., S.M., M.X., D.S., G.T.A., K.R.K., T.M.B.; Writing—review and editing: All authors; Visualization: C.P.; Final approval of manuscript: All authors; Funding acquisition: L.L.R., G.T.A.

Data availability

The Childhood Cancer Survivor Study is a US National Cancer Institute-funded resource (U24 CA55727) to promote and facilitate research among long-term survivors of cancer diagnosed during childhood and adolescence. CCSS data are publicly available on dbGaP at https://www.ncbi.nlm.nih.gov/gap/ through its accession number phs001327.v2.p1. and on the St Jude Survivorship Portal within the St Jude Cloud at https://survivorship.stjude.cloud/. In addition, utilization of the CCSS data that leverages the expertise of CCSS Statistical and Survivorship research and resources will be considered on a case-by-case basis. For this utilization, research Application Of Intent followed by an Analysis Concept Proposal must be submitted for evaluation by the CCSS Publications Committee. Users interested in utilizing this resource are encouraged to visit http://ccss.stjude.org. Full analytical data sets associated with CCSS publications since January of 2023 are also available on the St Jude Survivorship Portal at https://viz.stjude.cloud/community/cancer-survivorship-community~4/publications.

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